

Jaskaran Dhiman, PhD, P.Eng.

Faculty Lecturer | Department of Bioresource Engineering

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PROFESSIONAL PROFILE

Bioresource engineer with academic appointments at McGill University and the University of Guelph. Research focuses on environmental sustainability, soil and water quality, hydrological and climate modelling, wastewater reuse, and data-driven methods for agricultural systems. **Experienced instructor** in computing, statistics, modelling, GIS, data analytics, and engineering science; **recognized with McGill's 2025 Macdonald Campus Award for Distinguished Teaching**.

RESEARCH INTERESTS

Sustainable agricultural and environmental systems; soil and water quality; wastewater reuse and remediation; biochar and soil amendments; hydrological and climate modelling; geospatial and remote-sensing applications; applied data analytics and machine learning for environmental prediction; engineering education.

EDUCATION

Degree	Field	Institution	Year
PhD	Bioresource Engineering	McGill University, Canada	2019
M.S.	Biosystems Engineering	Auburn University, USA	2014
B.Tech	Agricultural Engineering	Punjab Agricultural University, India	2009

ACADEMIC APPOINTMENTS

Position	Institution	Dates
Faculty Lecturer	Department of Bioresource Engineering, McGill University	Aug 2023–Present
Sessional Lecturer	School of Engineering, University of Guelph	Aug 2022–Dec 2022
Postdoctoral Researcher	School of Engineering, University of Guelph	Jan 2021–Nov 2022
Postdoctoral Researcher	Department of Bioresource Engineering, McGill University	Jan 2020–Jan 2021
Graduate Research Assistant (PhD)	Department of Bioresource Engineering, McGill University	Sep 2014–Dec 2019
Graduate Research Assistant (M.S.)	Department of Biosystems Engineering, Auburn University	Aug 2012–Aug 2014

ADDITIONAL PROFESSIONAL EXPERIENCE

Senior Data Scientist, Consumer Protection and Government Services, Government of Manitoba (Nov 2022–Aug 2023).

SELECTED HONOURS AND AWARDS

Year(s)	Award
2025	Teaching Excellence Award, McGill University
2018	Mitacs Globalink Research Award, \$6,000
2018	USDA-NIFA Student Travel Grant, \$1,600
2014–2019	Graduate Excellence Award, McGill University, \$34,300
2014–2019	Graduate Research Enhancement and Travel Award (GREAT), McGill University, \$2,100
2015–2018	Merit-based funding, Tri-Council Networks of Centres of Excellence, \$24,000
2014–2015	Merit-based funding, NSERC, \$8,000
2014–2015	Schulich Graduate Fellowship Entrance Award, McGill University, \$25,000

2012–2014	Graduate stipend and tuition support funded by the U.S. Department of Energy and U.S. Department of Agriculture, Auburn University
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GRANT AND PROPOSAL DEVELOPMENT EXPERIENCE

Contributed to interdisciplinary proposal development across Canada- and India-linked collaborations. Served as project manager on the Genome Canada LSARP-2020 proposal.

Awarded Funding

Agency	Project	Amount	Year
IC-IMPACTS	A Floating Treatment Wetland System for Removing Contaminants from Rivers and Streams using a Biomimicry Approach	\$20,000	2018
Mitacs	Development of biochar-based water filtration system for treating municipal and industrial wastewater entering rivers and streams	\$6,000	2018

Selected Proposal Development Contributions

- Genome Canada (LSARP-2020) — Development of a stakeholder-approved, proactive, and early-warning bio-surveillance expert system for monitoring river health using genomic auditing (\$6,000,000, 2021–2025; not awarded).
- NSERC — Microplastic and nanoplastic pollution of agricultural ecosystems from repeated applications of municipal biosolids in soil (\$750,000, 2021–2025; not awarded).
- NFRF – International — A One Health approach to investigate applications of reclaimed water and biosolids in agriculture (\$412,500, 2023–2026; not awarded).
- IC-IMPACTS / McGill University (MI4) — Development of guidelines to ensure safe drinking water in locked-down buildings post-COVID-19 pandemic (\$126,750, 2020–2021; not awarded).

PUBLICATIONS

Journal Articles (*n* = 28; Google h-index = 14 as of Apr 2026)

Listed below in reverse chronological order.

1. Panjabi K., R. Rudra, R. Shukla, **J. Dhiman**, P. Goel, and B. Gharabaghi. 2025. Variable source area hydrology: past, present and future: a review. *Environmental Reviews* 33, 1-9.
2. Purkayshtha, J., S. Prasher, M. Afzal, **J. Dhiman**, and C. Nzediegwu. 2025. Effect of biochar, hydrogel and soil liming on nutrient leaching in a coarse-textured soil. *Sustainability* 17 (12), 5396.
3. Khosravi, K., Farooque, A. A., Bateni, S. M., Jun, C., & Dhiman, J. (2025). Prediction of three vital rainfall characteristics: Advanced hybrid tree- or lazy-based learner? *Results in Engineering*. 25, 103840
4. Tokaldany, E., R. Rudra, P. Goel, S. Prasher, P. Daggupati and **J. Dhiman**. 2024. Impact of Climate Change on the Number of Frost-Free Winter Days in Prairies and Eastern Provinces of Canada. *Journal of the ASABE*. 67 (3), 797-808
5. Kaur B., **J. Dhiman**, R. Rudra, P. Daggupati, P. Goel, M. Macrae, S. Prasher, and R. Shukla. 2024. Evaluation of Winter Hydrology performance of three field-scale models. *Journal of the ASABE*. 67 (3), 809-822
6. Rudra R., R. Shukla, T. Dickinson, P. Goel and **J. Dhiman**. 2024. Effect of Temperature on Frost-Free Days and Length of Crop Growing Season across Southern Ontario. *American Journal of Climate Change*. 12(4) 700-718.
7. Allataifeh N., R. Rudra, P. Daggupati, **J. Dhiman**, P. Goel and S. Prasher. 2024. A Spatiotemporal Variability in the C-factor: An Analysis Using High Resolution Satellite Imagery. *International Journal of Sediment Research*. 39(1), 83-89.
8. Singh S.*, T. Naik*, A. Anil*, **J. Dhiman***, V. Kumar, D. Dhanjal, L. Aguilar-Marcelino, J. Singh, P. Ramamurthy. 2022. Micro (Nano) Plastics in Wastewater: A Critical Review on Toxicity Risk Assessment, Behaviour, Environmental Impact and Challenges. *Chemosphere*. 290, 133169. * Equal Contribution
9. Rudra R., T. Dickinson, **J. Dhiman**, S. Manzoor, P. Goel and R. Shukla. 2022. Exceedance Probability Model for Predicting the Frequency of Frost-Free Days. *Journal of the ASABE*. 65(6), 1249-1256.
10. Purkayshtha, J., S. Prasher, M. Afzal, C. Nzediegwu, and **J. Dhiman**. 2022. Wheat straw biochar amendment significantly reduces nutrient leaching and increases green pepper yield in a less fertile soil. *Environmental Technology & Innovation*. 28, 102655.
11. **Dhiman J.**, S. Prasher, E. ElSayed, C. Nzediegwu, A. Mawof and R. Patel. 2021. Use of Polyacrylamide Superabsorbent Polymers and Plantain Peel Biochar to Reduce Heavy Metal Mobility and Uptake by Wastewater Irrigated Potato Plants. *Transactions of the ASABE*, 63(1), 11-28.

12. **Dhiman J.**, S. Prasher, E. ElSayed, R. Patel, C. Nzediegwu, and A. Mawof. 2021. Effect of Hydrogel based Soil Amendments on Yield and Growth of Wastewater Irrigated Potato and Spinach Grown in a Sandy Soil. *Environmental Technology and Innovation*. 23, 101730.
13. Azarkhish A., R. Rudra, P. Daggupati, **J. Dhiman**, T. Dickinson and P. Goel. 2021. Investigation of long-term climate and streamflow patterns in Ontario. *American Journal of Climate Change*. 10(4), 467-489.
14. **Dhiman J.**, S. Prasher, E. ElSayed, C. Nzediegwu, R. Patel and A. Mawof. 2020. Heavy Metal Uptake by Wastewater Irrigated Potato Plants Grown on Contaminated Soil Treated with Hydrogel based Amendments. *Environmental Technology and Innovation*, 100952.
15. **Dhiman J.**, S. Prasher, E. ElSayed, R. Patel, C. Nzediegwu, and A. Mawof. 2020. Effect of Hydrogel Based Soil Amendments on Heavy Metal Uptake by Spinach Grown with Wastewater Irrigation. *Journal of Cleaner Production*. 311, 127644.
16. Guan, P., S. Prasher, M. Afzal, S. George, J. Ronholm, **J. Dhiman**, and R. Patel. 2020. Removal of Escherichia coli from lake water in a biochar-amended biosand filtering system. *Ecological Engineering*, 150, 105819.
17. Singh S., V. Kumar, S. Datta, D. Dhanjal, D. Bhatia, **J. Dhiman**, J. Samuel and R. Prasad. 2020. A sustainable paradigm of sewage sludge biochar: Valorization, opportunities, challenges and future prospects. *Journal of Cleaner Production*, 122259.
18. Zhou, T., S. Prasher, Z. Qi, S. George, A. Mawof, C. Nzediegwu, **J. Dhiman** and R. Patel. 2020. Impact of Silver Nanoparticles in Wastewater on Heavy Metal Transport in Soil and Uptake by Radish Plants. *Water, Air, and Soil Pollution*. 232(7), 267.
19. Nzediegwu, C., S. Prasher, E. ElSayed, **J. Dhiman**, A. Mawof & R. Patel. 2019. Impact of Soil Biochar Incorporation on the Uptake of Heavy Metals Present in Wastewater by Spinach Plants. *Water, Air and Soil Pollution*. 231(3), 1-19.
20. Nzediegwu, C., S. Prasher, E. ElSayed, **J. Dhiman**, A. Mawof & R. Patel. 2019. Biochar applied to soil under wastewater irrigation remained environmentally viable for the second season of potato cultivation. *Journal of Environmental Management*, 254, 109822.
21. Nzediegwu, C., S. Prasher, E. ElSayed, **J. Dhiman**, A. Mawof and R. Patel. 2019. Effect of biochar on Yield of Potatoes Cultivated Under Wastewater Irrigation for Two Seasons. *Journal of Soil Science and Plant Nutrition*, 1-13.
22. Nzediegwu, C., S. Prasher, E. ElSayed, **J. Dhiman**, A. Mawof and R. Patel. 2019. Effect of biochar on heavy metal accumulation in potatoes from wastewater irrigation. *Journal of Environmental Management*, 232, 153-164.
23. **Dhiman J.**, J.S. Dhiman, A. Kaur, R. Aggarwal and M. Dhiman. 2016. New Approaches and Strategies for Irrigation Water Management. *Indian Journal of Economics and Development*. 12(1):1-14.
24. **Dhiman J.**, O. Fasina, A. Shrestha, S. Adhikari, B. Via and T. Gallagher. 2015. Physical, Ignition and Volatilization Properties of Biomass Feedstock Dusts. *Transactions of ASABE* 58(6), 1425-1437.
25. **Dhiman J.**, J.S. Dhiman, R. Aggarwal and M. Dhiman. 2015. Water Management for Sustainability of Irrigated Agriculture: An Indian Perspective. *Indian Journal of Economics and Development*. 11(3): 111-122.
26. Dhiman M. and **J. Dhiman**. 2015. Infusion of Farm Mechanization Technologies in Indian Agriculture: Progress and Impact. *Indian Journal of Economics and Development*. 11(1):125-136.
27. **Dhiman J.**, M.P. Kaushal and S. Garg. 2011. Well Spacing in Central Indian Punjab: A Case Study. *Journal of Crop Improvement*, Taylor and Francis. Vol. 25 issue: 2. P 151-160.
28. **Dhiman J.**, R. Shukla, R. Rudra, P. Daggupati, S. Prasher and P. Goel. 2026. SWAT-TM: A Machine-Learning-Based Soil Temperature Module to Improve Winter Hydrology and Nutrient Simulation in a Great Lakes Agricultural Watershed. *Journal of Environmental Modelling and Software*, *Under review*. ENVSOFT-D-25-02780.

Additionally, four research papers are in preparation to be submitted soon.

Book Chapter

1. Singh, S., H. Kaur, H. Assefa, D.S. Dhanjal, **J. Dhiman**, J. Singh, P. Ramamurthy. 2025. Recent Biotechnological Interventions for Sustainable Treatment of Industrial Wastewater Following Membrane-based Methods. *Recent Trends in Industrial Wastewater Treatment* (CRC Press), 94-122.

Conference Proceedings

1. **Dhiman J.**, S.O. Prasher, A. Mannan, E ElSayed and C. Nzediegwu. 2015. Use of Superabsorbent Polymers (Hydrogels) to Promote Safe Use of Wastewater in Agriculture. *Canadian Society for Civil Engineering 22nd Hydrotechnical Conference*, Montreal, Canada. Conference paper. Apr 2015.
2. Nzediegwu C., S.O. Prasher, A. Mannan, E ElSayed and **J. Dhiman**. 2015. Role of Plantain Peel Biochar in Enhancing Safe Use of Untreated Wastewater. *Canadian Society for Civil Engineering 22nd Hydrotechnical Conference*, Montreal, Canada. Conference paper. Apr 2015.

Theses

1. **Dhiman J.** (2014). Prediction of Heating and Ignition Properties of Biomass Dusts Using Near Infrared Spectroscopy. M.S. Thesis – Department of Biosystems Engineering, Auburn University, Auburn, AL, USA.
2. **Dhiman J.** (2019). Use of Hydrogel based Soil Amendments to Promote Safe Use of Wastewater in Agriculture. PhD Thesis – Department of Bioresource Engineering, McGill University, Montreal, QC, Canada.

TEACHING EXPERIENCE

As faculty lecturer, I have taught / co-taught a total of **15 undergraduate courses** across a wide range of topics in engineering, statistics, computing, GIS, modelling, data analytics, physics, and scientific communication, with enrollments ranging from small graduate seminars to large classes of approximately 140 students.

Course	Title	Term(s)	Approx. Enrollment
ENGG 4360	Soil Water Conservation Systems	Fall 2022	~16
BREE 252	Computing for Engineers	Fall 2023, 2024, 2025	~40
BREE 312	Electrical Circuits and Machines (co-instructed)	Fall 2023	~30
AEMA 310	Statistical Methods I	Winter 2024, 2025, 2026	~140
BREE 501	Simulation and Modelling (co-instructed)	Winter 2024, 2025, 2026	~30
BINF 511	Bioinformatics for Genomics (co-instructed)	Winter 2024	~20
AEPH 112	Introductory Physics 1	Fall 2024	~101
AEPH 113	Physics 1 (engineering students)	Fall 2024	~13
BREE 214	Geomatics (co-instructed)	Fall 2024	~35
BREE / ENVB 529	GIS for Natural Resource Management (co-instructed)	Fall 2024, 2025	~108
AEMA 204	Data Analytics for Biosystems	Winter 2025, 2026	~40
BREE 451/452/453	UG Seminar / Poster / Scientific Writing (co-instructed)	Fall 2025	~80

Teaching Assistant, McGill University (2016–2019): Statistical Methods I; Computing for Engineers (MATLAB); Hydrology and Water Resources; Equipment Management; Mechanics of Materials. Responsibilities included laboratories, review sessions, assignment and exam grading, and student support in problem solving.

Designed AEMA 204, Data Analytics for Biosystems, as an introductory data-analysis course using open-source tools.

Experience includes course design, laboratory and experiential learning, large-class instruction, and graduate-attribute assessment aligned with CEAB expectations.

STUDENT MENTORING AND SUPERVISION

Mentored a **total of 17** undergraduate and graduate research and design projects spanning soil and water systems, hydrological modelling, climate impacts, machine learning, and environmental analytics. **Selected records:**

Student	Project / Topic	Period
Peining Guan	Biochar-amended biosand filtering system for Escherichia coli removal from lake water	2019
Joba Purkayshtha	Soil amendments, nutrient leaching, and green pepper yield/quality	2019–2020
Tianai Zhou	Silver nanoparticles in wastewater and heavy metal uptake by radish plants	2020
Akshara Chandrabalan	Predicting biochar sorption capacities using artificial neural networks	2020
Baljeet Kaur	Evaluation of field-scale models for Canadian winter hydrology	2021–2022
Nabil Allataifeh	Spatiotemporal variability in USLE C-factor using satellite imagery	2021–2022
Nicholas Hitsman	Freeze–thaw cycles, soil properties, and phosphorus dynamics	2021–2022
Kangxu He	Climate change effects on plant growth, herbicide degradation, and soil microbial communities	2024–2025
Mankaran Dhiman	Nanofertilizers, nutrient leaching, greenhouse gases, and crop response	2024–2025
Benjamin Reny	Surface hydrodynamic modelling for Southern Quebec: FloodIQ	2024–2025
Aleix Toews	Improvements for Kwanlin Dün First Nation’s salmon carcass survey	2025–2026

INVITED TALKS (*n* = 12)

- **Resilient Education Systems for an Uncertain Future.** Invited talk, EDUCON-2026 International Conference, RIMT University, India, Apr 11, 2026.
- **AI in Engineering,** for BREE 490 (Engineering Design 2) course, at McGill University, Canada. Mar 24, 2026.
- **A Brief History of Computing and Responsible Use of AI,** for ANSC 514 (Coding for Production Data) course, at McGill University, Canada. Nov 26, 2025.
- **Responsible Use of AI in Engineering,** for BREE 490 (Engineering Design 2) course, at McGill University, Canada. Nov 04, 2025.
- **Our Changing Climate: Impact on the Environment and Global Health.** Invited talk, Interprofessional Global Health Course, McGill University, Feb 18, 2025.
- **Invited Speaker, VADA Program Summer School Career Development Panel,** University of Manitoba, Jun 23, 2023.
- **How SWAT Simulates Hydrology: Some Concepts.** Guest lecture, ENGG 6800 Deterministic Hydrological Modelling, University of Guelph, Nov 23, 2021.
- **Delivered keynote address on sustainable solutions to tackle environmental challenges.** MDISET Global Conclave, CT University, India, Sep 6, 2020.
- **Concerns for Sustainability of Punjab Agriculture.** Online expert interface, CT University, India, Apr 29, 2020.
- **Quantifying the Contaminant Removal Effect of Biochar from Soil and Water Samples.** Laboratory workshop, PLNT 430 Pesticides in Agriculture, McGill University, Jan 22, 2020.
- **Role of Biochar in Agriculture. Special lecture,** BREE 533 Water Quality Management, McGill University, Oct 31, 2019.
- **Promoting Safe Use of Wastewater for Agriculture Using Hydrogels and Biochar.** Water: Challenges and Opportunities workshop, University of Alberta, Jul 24–25, 2017.

SELECTED CONFERENCE PRESENTATIONS

Author/co-author on **42 oral conference / university research event presentations** and **12 poster presentations**.

Selected presentations are listed below:

- J. Dhiman, R. Shukla, R. Rudra, S. Prasher, P. Daggupati, P. Goel. “Integrating Novel Soil Temperature Estimation Routine with SWAT for Improved Winter Hydrological and Nutrient Modeling for Lake Erie Sub-basin.” ASABE Annual International Meeting, Toronto, Canada, Jul 13–16, 2024. Paper #2501606.
- J. Dhiman, R. Shukla, R. Rudra, P. Daggupati, S. Prasher, P. Goel. “Enhancing SWAT Model for Improved Winter Hydrology Simulation in the Great Lake Regions.” ASABE Annual International Meeting, Anaheim, California, USA, Jul 28–31, 2024. Paper #2401340.
- J. Dhiman, R. Rudra, P. Daggupati, P.K. Goel, S. Prasher, R. Shukla, P. Paul. “Modification of SWAT model for improved winter hydrology simulation in the Great Lakes Basin.” International Association for Great Lakes Research, Toronto, Canada, May 11, 2023.
- J. Dhiman, R. Rudra, P. Daggupati, P.K. Goel, S. Prasher, R. Shukla, P. Paul. “Improvement in the SWAT model to Simulate Canadian Winter Hydrology.” ASABE Annual International Meeting, Houston, Texas, USA, Jul 19, 2022. Paper #2201041.
- J. Dhiman, S. Prasher, E. ElSayed, C. Nzediegwu, A. Mawof, P. Guan. “Can we use Wastewater to Grow Food?” ASABE Annual International Meeting, Boston, Massachusetts, USA, Jul 8, 2019. Paper #1901267.
- J. Dhiman. “Promoting Safe Use of Wastewater for Irrigation: Innovative Strategies.” Innovations in Sustainable Water Resource Management (ISWRM-2018), Ludhiana, India, Jul 2, 2018.
- J. Dhiman, S. Prasher, E. ElSayed, C. Nzediegwu, A. Mawof. “Effect of Super Absorbent Polymers and Plantain Peel Biochar on Yield of Potato Grown with Wastewater Irrigation.” ASABE Annual International Meeting, Detroit, Michigan, USA, Jul 29–Aug 1, 2018. Paper #1801735.
- J. Dhiman, O. Fasina, B.K. Via, S. Adhikari, T.P. McDonald. “Prediction of Heating and Ignition Risks of Biomass Dusts Using Near Infrared Spectroscopy.” ASABE Annual International Meeting, Montreal, Canada, Jul 14, 2014. Paper #1914277.

PROFESSIONAL SERVICE

- Graduate and Postdoctoral Mentor, McGill University Graduate and Postdoctoral Support Program (Dec 2023–Present)
- Outreach and Alumni Coordinator, Student Engagement Committee, IC-IMPACTS (Jan 2020–Present)
- Chair, Student Engagement Committee, IC-IMPACTS (2019–2020)
- Organizing Committee Member, Water and Environment Talks (WESTalks) speaker series (2020–2021)

- Honorary Member, International Advisory Board, CT University, India (2018–Present)
- Mitacs Globalink Mentor, Mitacs (Mar 2019–Oct 2019)
- Volunteer, Let's Talk Science (2018–2019)

Reviewer for peer-reviewed journals including: Industrial Crops and Products; Journal of Crop Improvement; Frontiers of Agricultural Science and Engineering; Chemical and Biochemical Engineering; Agronomy; Journal of Irrigation and Drainage of ICID.

PROFESSIONAL AFFILIATIONS

- Professional Engineers Ontario (PEO), Canada
- American Society of Agricultural and Biological Engineers (ASABE)
- Canadian Society for Bioengineering (CSBE)